

# 2022秋冬学期 海洋环境与气候变化 回忆卷

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## 基本信息

- 开课学院：海洋学院
- 学分：2
- 课程类型：国际化模块
- 考试形式：开卷，英文答题

## Q1 (30 points)

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- Explain the terms photosynthesis and respiration. (5)
- Explain the differences in the distribution of carbon dioxide near the surface and in the bottom water column, using the knowledges of biological pump and solubility pump. (10)
- What impacts does the climate change have on marine carbon cycle? (15)

## Q2(30 points)

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- What are the social impacts of floods? (5)
- What are the causes of the occurrence of floods? (10)
- Give examples of short-duration, high-discharge events. And what impacts do short-duration, high-discharge events have

on the water cycle near coastal areas? (15)

## Q3(40 points)

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- Explain the differences between autotrophy and heterotrophy. (4)
- What are the meanings of terms allochthonous and autochthonous? (4)
- What is the influence of increased allochthonous input on the autotrophic/heterotrophic areas. (8)
- What are the importances of mangrove and salt marshes ecosystems? (8)
- What are the threats to those ecosystems? (8)
- Give some suggestions on what we can do to protect the wetland ecosystems. (8)